Ship Detection in SAR Images Based on Multilevel Superpixel Segmentation and Fuzzy Fusion

Qian Sun, Ming Liu, Shichao Chen, Fugang Lu, and Mengdao Xing, *Fellow, IEEE*

***Abstract*— Superpixel can maintain the boundary of the target and reduce the influence of speckle noise, which has been widely applied to synthetic aperture radar (SAR) image target detection. But the size of the superpixel has a great impact on the performance of superpixel-based SAR target detection algorithms. To solve this problem, we propose a multilevel ship target detection algorithm based on superpixel segmentation. First, the SAR images are segmented in different levels with different superpixel sizes. Different descriptions of the SAR images are obtained in different levels. Second, we determine the feature of the superpixels in each level. And to enhance the adaptability of the proposed algorithm, we propose an adaptive distance calculation method to select the contrast superpixels in each level. Third, the soft detection results are realized in each level using the fuzzy *C*-means (FCM) algorithm. Finally, the soft detection results obtained in different levels are fused by a new fusion strategy to achieve the final ship target detection result. The influences caused by different superxiel sizes can be effectively eased by fusion. Experiments in different SAR images have verified the effectiveness of the proposed algorithm in accurately detecting ship targets and insensitivity to the superpixel size.**

***Index Terms*— Fuzzy fusion, multilevel superpixel segmentation, ship detection, synthetic aperture radar (SAR) images.**

I. Introduction

REMOTE sensing technology plays an important role in both civil and military fields. At present, remote sensing images mainly include infrared images, optical images, and synthetic aperture radar (SAR) images. Among them, SAR, as an active microwave imaging sensor, has many advantages over infrared, optical, and other passive imaging sensors [1], [2]. Its advantages are as follows: first, SAR uses active microwave imaging, which can penetrate the interference environment such as clouds, dense fog, and smoke and obtain observation images under the meteorological conditions with extremely poor visibility. Second, SAR can image the observation scene with high resolution, which is helpful to the interpretation and understanding of the observation image. Third, the resolution of SAR systems is independent of the observation distance and has the ability of long-distance detection. In short, compared with passive imaging sensors, SAR has been widely used in both civil and military fields because of its advantages of all-day, all-weather, and long-distance working ability [1], [2], [3], [4], [5], [6], [7], [8].

In both the civil and military fields, ship target detection is an important application of SAR [8], [9]. In the civil field, ship target detection can provide information support for ship navigation in busy sea areas, ship entry and exit management, fishery monitoring, ship oil spill pollution detection, and other activities. In the military field, ship target detection can locate and track the port and sea targets, maintain the territorial sea order in peacetime, and defeat the enemy in wartime.

A good target detection algorithm should be fast and accurate, and the detected targets should be complete. Superpixels are small areas consisted of multiple pixels with adjacent positions and similar characteristics such as color, brightness, and texture [1]. Most superpixels can well preserve the boundary and structure information of objects in the image [2], [8]. Moreover, since the number of superpixels is far less than the number of pixels, the processing time can be greatly reduced [8]. Therefore, several superpixel-based target detection algorithms have been proposed. However, the selection of superpixel size has a strong impact on the final detection result, especially when the ship target size in SAR image has a strong fluctuation.

We propose a new ship target detection algorithm based on multilevel superpixel segmentation and fuzzy fusion to ease the influence of superpixel size selection in this article. First, the superpixels of different levels are obtained by the superpixel segmentation. Second, the superpixel features of different levels are adaptively extracted. Third, the soft detection results of different levels are made using the obtained superpixel features. Finally, the ship target detection result is acquired by the fuzzy fusion of multilevel soft detection results. The main contributions and novelties of the proposed algorithm are threefolds.

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A. Multilevel Fusion

The proposed algorithm uses different superpixel sizes to get the multilevel superpixel segmentation results. The superpixel segmentation results in different levels can provide different feature information. The proposed algorithm makes use of the different feature information by multilevel fusion to improve the performance of ship target detection and weaken the influence of superpixel size selection.

B. New Method of Selecting Contrast Superpixels

We establish a new method to adaptively calculate the distance of the test superpixel and its contrast superpixels, where the test superpixel is the superpixel whose feature is being extracted, and the contrast superpixels are essentially the background superpixels. Using the adaptive distance, we adaptively select the contrast superpixels which make the calculated superpixel features more obvious in different levels. The new method of selecting the contrast superpixels makes the proposed algorithm more adaptive.

C. New Fusion Strategy

In the fusion process, we propose a new fusion strategy. The strategy fuses different features obtained in different levels, which makes the proposed algorithm more robust. Using the proposed strategy, the soft detection results of different levels are gradually fused together to obtain a good ship target detection result.

The rest of this article is arranged as follows. In Section II, we introduce some related work. In Section III, we describe the proposed algorithm in detail. In Section IV, the proposed algorithm is tested and compared with other algorithms. Finally, we summarize this article in Section V.

II. Related Work

A. SAR Ship Detection Algorithms

For SAR ship target detection, due to the difference in microwave scattering characteristics between the ship and the surrounding sea surface, the backscattering of the ship is stronger than that of the sea surface [10]. Therefore, the ship target can be detected by setting an appropriate threshold [5]. The widely used constant false alarm rate (CFAR) algorithm compares the pixel intensity and the threshold to achieve ship target detection [11]. However, with the progress of remote sensing technologies, the resolution of SAR images becomes higher and higher. Higher resolution images can provide more detailed target information and also bring challenges to target detection in SAR images. Traditional CFAR target detection algorithms, such as the two-parameter CFAR detector proposed by Lincoln Laboratory [4], [8], have three obvious defects when processing high-resolution SAR images. First, since the echo of the target exists in multiple units [9], the pixel intensity value of the target will fluctuate up and down within a certain range. When using the traditional CFAR algorithms to detect the target, some low-intensity pixels of the target may be missed, and the final detected target will be broken or discontinuous. Second, in high-resolution SAR images, speckle noise is more obvious [8], which leads to more false alarms in the detection results. Third, traditional CFAR target detection algorithms traverse all the pixels of SAR image, which is very time-consuming [5], [8].

Due to the above defects, researchers have made a lot of improvements to the traditional CFAR target detection algorithm. For example, Leng et al. [12] thought that the spatial distribution is as important as the intensity distribution, so they generated a combined image by combining the intensity distribution and the spatial distribution of the SAR image, and then applied a standard CFAR detection algorithm to detect targets. Ai et al. [13] believed that clutter samples are often polluted by outliers in complex environments. Therefore, to improve the detection performance, a CFAR target detection algorithm based on bilateral clipping statistics was proposed. At the same time, researchers also proposed some non-CFAR target detection algorithms. For instance, Wang et al. [14] used the tail scattering characteristics of ship targets for target detection. Song et al. [15] extracted the sparse features of ship targets in SAR images, and then transformed the ship detection problem into a sparse signal recovery problem. Wang et al. [16] used Markov random field to extract intraimage saliency map and used fuzzy C-means (FCM) to construct interimage saliency map, and finally fused intraimage saliency map and inter-image saliency map to achieve target detection. However, all the above target detection algorithms are based on pixels, but in fact, ship targets are usually formed by irregular connected domains in SAR images. During the detection process, the pixel-based algorithms may lose target pixels with weak intensity. The detected target will be fractured or even lost. And the bad results eventually will perplex the subsequent recognition and classification process.

To solve the above problems, some superpixel-based algorithms have been proposed. For instance, Pappas et al. [9] used superpixels instead of rectangular sliding windows to define the protection area and the clutter area in CFAR algorithms, which reduced the false alarm rate. Li et al. [7] used the Bhattacharyya distance as the basis to derive the superpixel statistical dissimilarity, established the ship target–clutter contrast enhancement map, and verified the performance of target detection in the port area and the nearshore sea area. Wang et al. [2] proposed a superpixel-based detector, which used multiorder superpixel features based on the Gaussian mixture model to distinguish the ship targets and the sea clutter. Sun et al. [8] proposed a two-stage and superpixel-based CFAR target detection algorithm using the statistical characteristics in different scenes.

In some ship detection tasks, to improve the accuracy of the target detection, some researchers have adopted multisource fusion methods. For instance, Zhu et al. [17] fused spaceborne and airborne SAR images based on multiorder superpixel-level saliency and fuzzy logic to realize ship target detection. Liu et al. [18] first fused the optical data and SAR data to obtain the port slice image containing ship targets, and then used

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different characteristics of the data source to extract the ship target detection results.

B. Superpixel Segmentation Algorithms

Superpixel segmentation algorithms can be divided into two categories, the first is the graph-based algorithms, and the second is the gradient-ascent-based algorithms [19]. The graph-based algorithms take the pixels in the image as nodes, map the image into a weighted undirected graph, and transform the image segmentation problem into a function optimization problem by constructing a loss function to realize superpixel segmentation. The graph-based superpixel segmentation algorithms include graph cuts [20], superpixel lattices [21], normalized cuts (NCs) [22], etc. The algorithms based on gradient ascent use the iterative idea. After initializing the clustering, the gradient ascent algorithms repeatedly update until the error of the clustering results between two adjacent iterations meets the requirement. Superpixel segmentation algorithms based on gradient ascent include Watershed [23], Turbopixel [24], mean shift [25], quick shift [26], simple linear iterative clustering (SLIC) [27], etc. Compared with other superpixel segmentation algorithms, SLIC has four main advantages. First, the segmented superpixels are as compact and tidy as cells, and the neighborhood features are easier to be expressed. Second, SLIC can not only process color images but also process grayscale images, e.g., SAR images. Third, there are very few parameters that need to be set. Fourth, it is ideal in terms of running speed, compactness of segmented superpixels, and contour retention [27].

The superpixel segmentation algorithms mentioned above were originally proposed for optical images. Some superpixel segmentation algorithms for SAR images have also been proposed, such as mixture-based superpixel (MISP) [28], edge-aware superpixel generation with one iteration merging (ESOM) [29], generalized gamma mixture model superpixel (MISP-GGD) [30], and edge penalty and contraction–expansion search strategy (EPCES) [31]. Moreover, in view of the characteristics of SAR images, many variants of the above algorithms have also been proposed, such as the pixel intensity and location similarity (PILS) [32], SLIC-generalized likelihood ratio (GLR) [7], SLIC-Mahalanobis distance (MD) [33], and Fisher vector (FV)-based adaptive superpixel segmentation (FVASS) [34].

C. Statistical Distance

In statistics, statistical distance quantifies the difference between two objects, in which the object can be a random variable, a probability distribution or a sample, or the distance between a single sample point and a population or larger sample point [35]. At present, common statistical distances include Rényi divergence [35], Jensen–Shannon (JS) divergence [36], Bhattacharyya distance [7], Kullback–Leibler (KL) divergence [37], Hellinger distance [35], total variation distance [38], and Wasserstein distance [39], where KL divergence, Hellinger distance, and total variation distance are special cases of f-divergence [40]. The Wasserstein distance is used to represent the distance between two distributions in this article. Compared with other distances, the first Wasserstein (w_1) distance [41] can well measure the difference between two discrete distributions. More importantly, the amount of data in the two distributions does not need to be the same when calculating the w_1 distance.

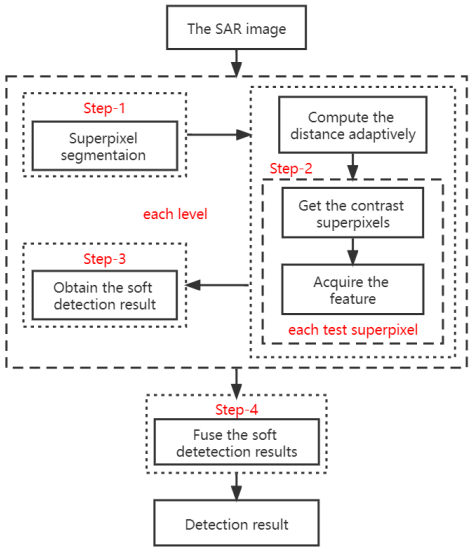


Fig. 1. Flowchart of the proposed algorithm.

III. Proposed Ship Detection Algorithm

The overall idea of the proposed algorithm is to achieve the extraction of SAR image features in different levels with different superpixel sizes at first, and then make soft detection results at each level, and finally fuse the soft detection results of each level to obtain the SAR ship target detection result. The flowchart of the proposed algorithm is shown in Fig. 1. The proposed algorithm mainly includes four steps: superpixel segmentation, acquisition of ship target detection features, obtaining soft detection results within each level, and fusion of different levels. In this section, we will discuss the four steps in detail.

A. Step-1: Superpixel Segmentation

Due to the focus on target detection after superpixel segmentation and the excellent performance of the SLIC algorithm, which is simple, fast, and works well enough, like in [1], [2], [5], and [9], we choose the SLIC superpixel segmentation algorithm to segment superpixels in this article. When the SLIC algorithm is used to segment superpixels in SAR images, the distance metric $D(i, j)$ between pixel i and pixel j can be defined as [27]

$$D(i,j)=\sqrt{\left(l_j-l_i\right)^2+\left(\frac{\lambda}{S}\right)^2\left[\left(x_j-x_i\right)^2+\left(y_j-y_i\right)^2\right]} \tag{1}$$

where l_i and l_j are the pixel values of pixel i and pixel j in the SAR image, respectively, which are called intensity in this article, (x_i, y_i) and (x_j, y_j) are the spatial coordinates of pixel i and pixel j in the SAR image, respectively, S is the preset

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Algorithm 1 SLIC for SAR Images

Input: The SAR image, superpixel size S, compactness parameter λ.

Step-I: Initialize cluster centers and evenly distribute seeds in the SAR image according to the superpixel size S.

Step-II: In the 3×3 neighborhood of the initial cluster centers, the gradient values of all pixels are calculated, and the pixels with the lowest gradient are reselected as the initial cluster centers.

Step-III: In the $2S \times 2S$ neighborhood of each cluster center, the distance between each pixel in the neighborhood and the cluster center is calculated using the distance metric in (1), and then the C-means [42] algorithm is used to connect each pixel with its nearest cluster center.

Step-IV: Calculate the average coordinates of all pixels of each cluster and update the cluster centers.

Step-V: Repeat Step-III and Step-IV until the cluster centers no longer change significantly.

Step-VI: Assign discontinuous superpixels or superpixels with too small size to adjacent superpixels to enhance connectivity.

Output: Superpixels of the SAR image.

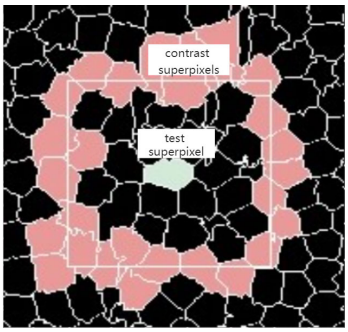


Fig. 2. Schematic of the test superpixel (light green) and its contrast superpixels (light coral).

superpixel size, and λ is the compactness parameter, which is used to balance the spatial distance and the intensity distance. The overall process of the SLIC algorithm for SAR images is shown in Algorithm 1 [27].

B. Step-2: Acquisition of Ship Target Detection Features

In this step, we adaptively select the contrast superpixels at first. And then, we calculate the distribution distances of the test superpixels and their contrast superpixels. Finally, the features of the superpixels are obtained.

1) Adaptive Selection of Contrast Superpixels: Since the size of the superpixels is different at different levels, the distance between the test superpixels and the contrast superpixels should be dynamically determined to enhance the adaptiveness of the detection algorithm. As shown in Fig. 2, we call the superpixel whose feature is calculated in the center as the test superpixel, and call the superpixels that the square passes through as the test superpixel's contrast superpixels.

For the determination of the distribution distance between the superpixels and their contrast superpixels, the distance

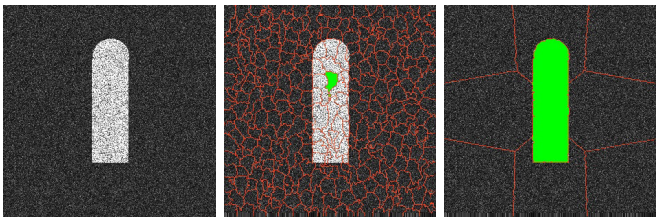


Fig. 3. (a) Simulated SAR image. (b) Superpixel segmentation of (a) with a small superpixel size. (c) Superpixel segmentation of (a) with a large superpixel size.

should be large when the superpixel size is small. And when the superpixel size is large, the distance should be small. We take an example to show this criterion clearly. Fig. 3(a) is a simulated SAR image. Fig. 3(b) is the superpixel segmentation result when the superpixel size is small, i.e., the lower level. Fig. 3(c) is the superpixel segmentation result when the superpixel size is large, i.e., the higher level. The superpixels marked in Fig. 3(b) and (c) are the test superpixels. In Fig. 3(b), if the superpixels adjacent to the test superpixel are taken as its contrast superpixels, the test superpixel will have the same category as its contrast superpixels. The feature of the test superpixel will not be obvious. So the superpixels that are farther from the test superpixel should be selected as its contrast superpixels. In Fig. 3(c), the superpixels adjacent to the test superpixel can be taken as its contrast superpixels.

To adaptively obtain the contrast superpixels, we define the adaptive distance between the test superpixel and its contrast superpixels as

$$d = \rho \cdot S_{\text{cur}} \tag{2}$$

$$\rho = \tau - \frac{S_{\text{cur}} - S_{\min}}{S_{\max} - S_{\min}} \cdot (\tau - 1) \tag{3}$$

where ρ is the distance coefficient, S_{cur} is the superpixel size of the current level, τ is the total number of the superpixel levels, $S_{\max}$ is the superpixel size of the highest level, and $S_{\min}$ is the superpixel size of the lowest level. It can be seen from (3) that the distance coefficient ρ changes with the superpixel size of the current level S_{cur}. When the superpixel size of the current level S_{cur} becomes larger, the distance coefficient ρ becomes smaller, and when the superpixel size of the current level S_{cur} becomes smaller, the distance coefficient ρ becomes larger. Thus, the distance d between the test superpixels and their contrast superpixels at different levels is adaptively calculated. And the contrast superpixels corresponding to the test superpixels are adaptively acquired at different levels.

2) Calculation of the Wasserstein Distance: After adaptively obtaining the contrast superpixels, we calculate the w_1 distance of the test superpixel and its contrast superpixels according to (4) to measure their differences. The w_1 distance of the test superpixel and its contrast superpixels is defined as [41], [43]

$$w_1(f_t, f_{c_i}) = \int_{-\infty}^{+\infty} |F_t - F_{c_i}| \tag{4}$$

where F_t and F_{c_i} are the cumulative distributions of the distributions f_t and f_{c_i}, respectively, f_t is the distribution of

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the test superpixel, and f_{c_i} is the distribution of the ith contrast superpixel corresponding to the test superpixel.

To reduce the false alarms, we take the median of all w_1 distances between the test superpixel and its contrast superpixels as the feature of the test superpixel [2].

C. Step-3: Obtaining Soft Detection Results Within Each Level

In this step, we calculate the soft detection result, i.e., the clutter probability of each pixel.

To acquire the soft detection results within each level, we use the FCM [44] algorithm to get the probabilities of the superpixels belonging to different categories at first. And then the probability category of superpixels is determined. Finally, the superpixel-level clutter probability to the pixel-level probability is converted to obtain the clutter probability of the pixels. The detailed process is given as below.

We take the superpixels' features calculated by Step-2 as the sample data, set the number of the categories to be c, cluster the samples using (5) and (6) of the FCM algorithm, and obtain the probability of superpixels belonging to different categories.

The iterative formulas of the FCM algorithm are defined as [44], [45]

$$u_{ij} = \frac{1}{\sum_{k=1}^{c} \left(\frac{d_{ij}}{d_{kj}}\right)^{\frac{2}{m-1}}} \tag{5}$$

$$c_i = \frac{\sum_{j=1}^{L} {u_{ij}}^m w_{1_j}}{\sum_{j=1}^{L} {u_{ij}}^m} \tag{6}$$

where u_{ij} is the probability that the jth superpixel belongs to category i, c_i represents the cluster center of category i, m is the fuzziness index (generally 2), d_{ij} is the Euclidean distance from the jth superpixel to the cluster center of category i, L is the number of superpixels, and w_{1_j} is the feature of the jth superpixel.

In this article, the number of the categories c is set to 2, which are the target and the clutter, respectively. According to the definition of the FCM algorithm, the relationship between the probabilities of different categories of superpixels is given as follows [44]:

$$\sum_{i=1}^{c} u_{ij} = 1 \quad \forall j = 1, \ldots, L \tag{7}$$

where the sum of the probabilities representing the same superpixel belonging to different categories is 1.

In real SAR images, there are two situations. The first is that the clutter area is larger than the target area, which is common. The second is that the target area is larger than the clutter area. According to the probabilities of different categories of superpixels, all the superpixels can be divided into two categories. Less superpixels correspond to the category of smaller area, whereas more superpixels correspond to the category of larger area [4]. Thus, the probabilities' category of superpixels can be determined.

The above probabilities are at the superpixel level. It is necessary to convert the superpixel-level probability into the pixel-level probability. In this article, we assign the clutter probability of each superpixel to the pixels it contains. And then the pixel-level clutter category probability matrix $\mathbf{Q}$ can be obtained. $\mathbf{Q}$ is a matrix that represents the probability of each pixel belonging to clutter.

TABLE I
PIXEL PROBABILITY FUSION BETWEEN DIFFERENT LEVELS

	Previous level	Current level
Case 1	Clutter probability	Clutter probability
Case 2	Clutter probability	Target probability
Case 3	Target probability	Clutter probability
Case 4	Target probability	Target probability

D. Step-4: Fusion of Different Levels

In this step, to improve the performance and robustness of the proposed algorithm, we fuse pixel clutter probabilities between different levels using a new fusion strategy.

At different levels, each pixel has two probabilities, namely, clutter probability and target probability. And there are four cases when fusing between different levels, as shown in Table I. In the four cases, the first three cases include clutter probability. The first three cases can be divided into two parts. The first part is Case 1. In Case 1, the calculated probability is determined as the clutter probability. And the second part includes Case 2 and Case 3. In Case 2 and Case 3, the calculated probability is determined as the uncertain probability.

Therefore, we construct the fusion strategy of clutter probability as

$${p_{ij}}^{(l)} = {p_{ij}}^{(l-1)} q_{ij} + \sigma_{ij} \tag{8}$$

$$\sigma_{ij} = \omega\left[{p_{ij}}^{(l-1)}\left(1 - q_{ij}\right) + q_{ij}\left(1 - {p_{ij}}^{(l-1)}\right)\right] \tag{9}$$

$$\omega = \frac{\sum_{i=1}^{r}\sum_{j=1}^{t} {p_{ij}}^{(l-1)} + \sum_{i=1}^{r}\sum_{j=1}^{t} q_{ij}}{2N}$$
$$\forall {p_{ij}}^{(l-1)} \in \mathbf{P}^{(l-1)} \quad \forall q_{ij} \in \mathbf{Q} \tag{10}$$

where ${p_{ij}}^{(l)}$ is the clutter probability of the pixel after fusion at the current level, ${p_{ij}}^{(l-1)}$ is the clutter probability of the pixel after fusion at the previous level, q_{ij} is the clutter probability of the pixel at the current level, σ_{ij} is the supplementary clutter probability of the pixel, ω is the clutter determination coefficient, N is the number of pixels of the SAR image, r is the number of rows of the SAR image, t is the number of columns of the SAR image, and $\mathbf{P}^{(l-1)}$ is the pixel-level clutter probability matrix after fusion at the previous level.

It can be seen from (8) that calculating ${p_{ij}}^{(l)}$ consists of two parts, corresponding to the first part and the second part of the first three cases, respectively. To determine the second part, we establish an adaptive clutter determination coefficient ω in (10). The criterion is: If the region of the SAR image is mostly covered by clutter, ω should be large, since the category of the pixel is more likely to be clutter, and its probability should be large. Similarly, if the region of the SAR image is mostly covered by targets, ω should be small, since the category of the pixel is more likely to be targets, and its probability should be small.

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Algorithm 2 Proposed Algorithm

Input: The SAR image, superpixel size of the lowest level S_{min}, superpixel size of the highest level S_{max}, total number of the superpixel levels τ.

Step-I: Set the compactness parameter λ, and take the superpixel size of the current level as the superpixel size S, the superpixels of the SAR image are segmented using Algorithm 1. The superpixel size starts with S_{min}, and the superpixel size is updated by Step-V.

Step-II: Select the contrast superpixels according to (2) and (3), then calculate the w_1 distances between the test superpixels and their contrast superpixels according to (4), and take the median as the feature of the superpixels.

Step-III: Take the superpixel features obtained by Step-II as the sample data, cluster the superpixels using the FCM algorithm, and obtain the pixel-level clutter probability matrix.

Step-IV: Fuse the pixel clutter probability according to (8), (9), and (10).

Step-V: Update the superpixel size, repeat Step-I, Step-II, Step-III, and Step-IV until the superpixel levels are out of range, then obtain the pixel level target probability matrix according to the pixel level clutter probability matrix, compare the two category probabilities of each pixel, select the category with higher probability as the final category of the pixel, and then obtain the ship target detection result image.

Output: Ship target detection result image.

The specific process of the proposed algorithm is shown in Algorithm 2.

IV. Experimental Results and Analysis

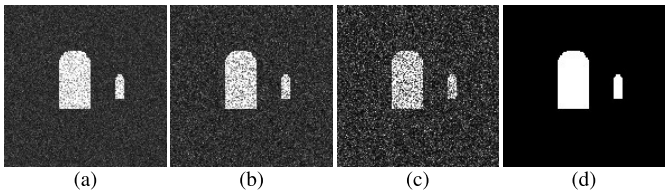
In this section, we compare the ship target detection performance of the proposed algorithm with two competitors, which are a bilateral-trimmed-statistics-based CFAR detector (BTS-RCFAR) [13] and a superpixel-based ship detector via the local contrast of FVs (LCFV-based detector) [2], respectively. And we conduct the experiments of ship target detection in the simulated SAR images and real SAR images. All the SAR images are common, i.e., the clutter area is larger than the target area.

To quantitatively compare the performance of the proposed algorithm and the two competitive algorithms, we compare the target detection accuracy (P_{dt}) and the target false alarm probability (P_{fa}) in pixel level [46], which can be expressed as

$$P_{\text{dt}} = \frac{N_{\text{td}}}{N_{\text{ground_truth}}} \tag{11}$$

$$P_{\text{fa}} = \frac{N_{\text{fd}}}{N_{\text{total_target}}} \tag{12}$$

where N_{td} is the number of correctly detected ship target pixels, $N_{\text{ground_truth}}$ is the number of pixels of the ship target in the ground truth, N_{fd} is the number of ship target pixels detected incorrectly, and $N_{\text{total_target}}$ is the number of pixels of all the ship targets in the detection result.



(a) (b) (c) (d)

Fig. 4. Simulated SAR images with Gamma noise. (a) Mean value of 1, variance of 0.1. (b) Mean value of 1, variance of 0.2. (c) Mean value of 1, variance of 0.5. (d) Ground truth.

In practical application, a good target detection method should make a compromise between detection accuracy and false alarm probability, so we use $F1$ score defined in (13) to evaluate the overall performance of the three algorithms. The maximum $F1$ score is 1 and the minimum is 0. The higher the score, the better the algorithm performance

$$F1 = 2 \times \frac{P_{\text{dt}} \times (1 - P_{\text{fa}})}{P_{\text{dt}} + (1 - P_{\text{fa}})}. \tag{13}$$

When the superpixel size is set to be too small or too large, the superpixels cannot reflect the boundary and structure information of the object. For instance, if the superpixel size is too small, a large number of false alarms may occur in the lowest level. Therefore, to avoid too many false alarms, the superpixel size of the lowest level S_{min} starts from 6 in the proposed algorithm. And we suggest that the superpixel size of the highest level S_{max} should not be greater than one third of the average ship target size. For convenience, we use two superpixel size combinations to verify the target detection performance of the proposed algorithm in this article. The two superpixel size combinations are 6, 12, 18, 24 and 8, 14, 20, respectively. And in the experiments, the compactness parameter λ of the superpixel segmentation is set to be 1.8 times of the superpixel size S. When the LCFV-based detector [2] detects the targets, the superpixel size is set to 6, 16, and 24, respectively. Other parameters of the LCFV-based detector are consistent with those suggested by Wang et al. [2]. The CFAR of BTS-RCFAR is 0.1, and other parameters are the same as [13].

A. Detection Results of Simulated Images

Three 145 × 145 simulated images [see Fig. 4(a)–(c)] are used to verify the effectiveness of the proposed algorithm in reducing the influence of superpixel size selection and the robustness in ship size fluctuation. In Fig. 4(a)–(c), the size of the large ship is about ten times of the size of the small ship. And the ship size refers to the number of the pixels occupied by the ship target.

By observing the results, as shown in Fig. 5(a)–(f), we can find that the proposed algorithm achieves satisfying detection results in simulated images. Although the detection results of the proposed algorithm become worse when the noise turns stronger, both large and small ship targets can be detected. When the superpixel size is 6, scattered parts of the ship targets are detected by the LCFV-based detector [2], as shown in Fig. 5(g)–(i). When the superpixel size is 16, the LCFV-based detector [2] can detect the ship targets in Fig. 4(a), but

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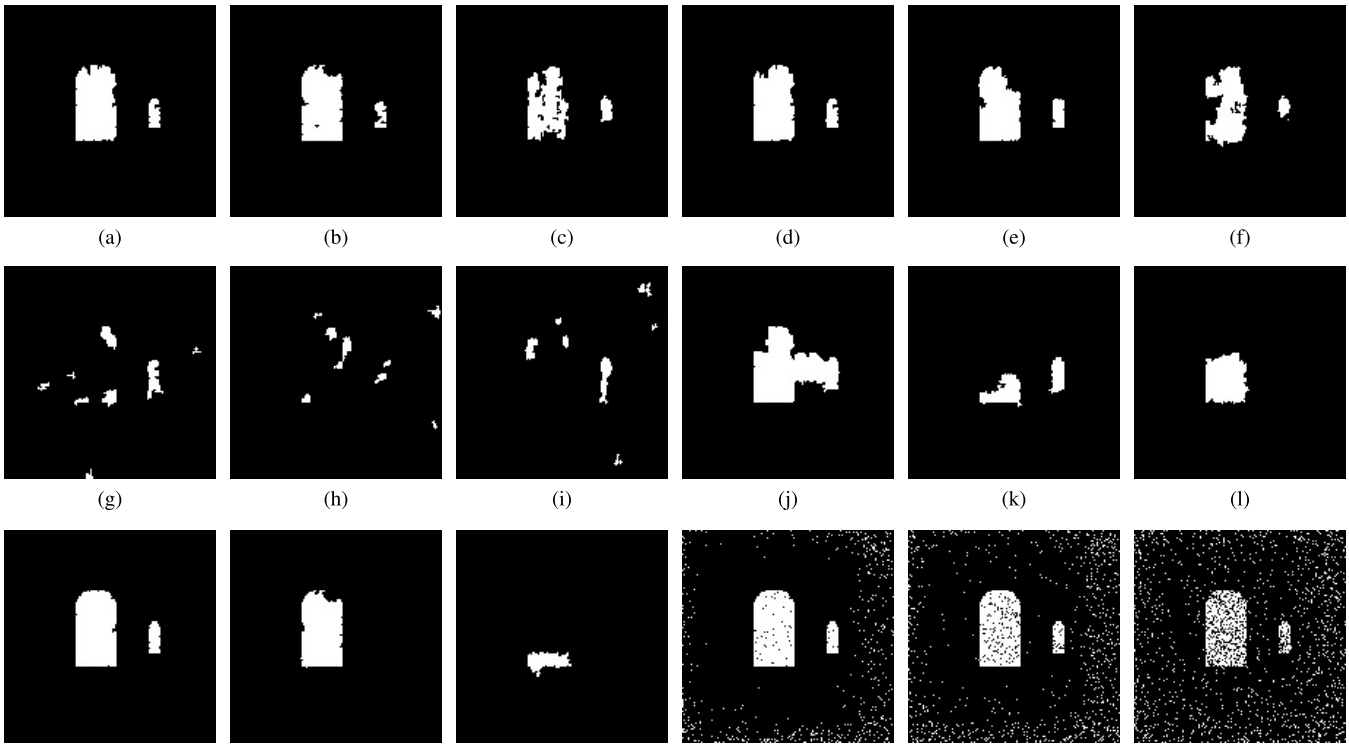


Fig. 5. Detection results of the simulated SAR images. (a)–(c) Detection results of Fig. 4(a)–(c) obtained by the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24. (d)–(f) Detection results of Fig. 4(a)–(c) obtained by the proposed algorithm when the superpixel sizes are 8, 14, and 20. (g)–(i) Detection results of Fig. 4(a)–(c) obtained by the LCFV-based detector [2] when the superpixel size is 6. (j)–(l) Detection results of Fig. 4(a)–(c) obtained by the LCFV-based detector [2] when the superpixel size is 16. (m)–(o) Detection results of Fig. 4(a)–(c) obtained by the LCFV-based detector [2] when the superpixel size is 24. (p)–(r) Detection results of Fig. 4(a)–(c) obtained by BTS-RCFAR [13].

there are a lot of false alarms and the two ship targets are connected, as shown in Fig. 5(j). LCFV-based detector [2] detects both the ship targets in Fig. 4(b), but only a small fraction of the large ship target is detected, as shown in Fig. 5(k). And the LCFV-based detector [2] only detects a part of the large ship target in Fig. 4(c), it fails to detect the small target, as shown in Fig. 5(l). When the superpixel size is 24, the LCFV-based detector [2] obtains the best detection result of Fig. 4(a), as shown in Fig. 5(m). But for the images shown in Fig. 4(b) and (c), the LCFV-based detector [2] only detects the large ship target or part of the large ship target, as shown in Fig. 5(n) and (o). BTS-RCFAR [13] detects both the ship targets in every simulated image, as shown in Fig. 5(p)–(r), but there are a lot of false alarms.

In the LCFV-based detector [2], the features of the superpixels are obtained by comparing adjacent superpixels. When the superpixel size is small relative to the ship target, the features of the superpixels are probably not obvious. Therefore, when the superpixel size is 6, the performance of the LCFV-based detector [2] is poor in detecting the large ship target and relatively good in detecting the small ship target, as shown in Fig. 5(g)–(i). And when the superpixel size is 16 or 24, the small ship target is missed, as shown in Fig. 5(l), (n), and (o). Due to the adaptive distance calculation within the level and the fusion of soft detection results among different levels, the proposed algorithm can achieve satisfying detection results with strong fluctuations of the ship target size. Since BTS-RCFAR [13] is a pixel-based algorithm, the detection results encounter noise problem, which can be effectively eased by the proposed superpixel-based algorithm.

Quantitatively, as shown in Table II, compared with the LCFV-based detector [2] and BTS-RCFAR [13], the proposed algorithm achieves higher P_{dt} and lower P_{fa} in general. Therefore, $F1$ score of the proposed algorithm is the highest, i.e., the overall detection performance of the proposed algorithm is the best. As can be seen from Table II, the detection performance of the LCFV-based detector [2] changes dramatically with different choices of the superpixel size.

From the detection results in Fig. 5 and the quantitative results in Table II, it can be concluded that the proposed algorithm is superior in reducing false alarms and accurately detecting ship targets. Especially, we can tell that the proposed algorithm greatly weakens the impact of superpixel size selection on the detection results.

B. Detection Results of Real Images

The real SAR images include two GaoFen-3 SAR images [see Figs. 6(a) and 7(a)], a TerraSAR-X SAR image [see Fig. 8(a)], and a Sentinel-1 SAR image [see Fig. 9(a)]. The detailed information of the real SAR images is shown in Table III. The Gaofen-3 images in Figs. 6(a) and 7(a) contain one ship target and four ship targets, respectively [47]. The TerraSAR-X image in Fig. 8(a) is obtained from the Singapore

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TABLE II
COMPARISON OF THE SHIP TARGET DETECTION PERFORMANCE OF THREE ALGORITHMS IN SIMULATED SAR IMAGES

Algorithms	Superpixel Sizes	Fig. 4(a)			Fig. 4(b)			Fig. 4(c)		
		P_{dt}	P_{fa}	$F1$	P_{dt}	P_{fa}	$F1$	P_{dt}	P_{fa}	$F1$
The Proposed Algorithm	6, 12, 18, 24	0.9330	**0.0**	0.9653	**0.8914**	**0.0**	**0.9426**	0.6858	**0.0**	0.8136
	8, 14, 20	0.9132	0.0014	0.9540	0.8519	0.0022	0.9190	**0.7011**	0.0188	**0.8179**
LCFV-Based Detector [2]	6	0.2043	0.2610	0.3202	0.0837	0.5019	0.1432	0.1373	0.3750	0.2251
	16	0.8921	0.2299	0.8266	0.3282	0.0586	0.4867	0.5294	0.0072	0.6905
	24	**0.9860**	**0.0**	**0.9929**	0.8372	0.0023	0.9104	0.1564	0.1343	0.2650
BTS-RCFAR [13]	-	0.9674	0.2395	0.8516	0.8876	0.3532	0.7483	0.6928	0.5201	0.5670

TABLE III
SPECIFIC INFORMATION OF THE REAL SAR IMAGES USED IN THE EXPERIMENTS

	Fig. 6(a) and Fig. 7(a)	Fig. 8(a)	Fig. 9(a)
Sensor	GaoFen-3	TerraSAR-X	Sentinel-1
Frequency Band	C	X	C
Centre Frequency	5.4GHz	9.65GHz	5.405GHz
Imaging Mode	Ultrafine Strip-Map (UFS)	StripMap (SM)	Interferometric Wide-Swath (IW)
Polarization	HH	HH	VH
Resolution	3m×3m	7.14m×7.15m	5m×20m
Swath	30km	30km	250km
Incident Angle	20°∼50°	22.24°∼25.59°	27.6°∼34.8°
Pixels	256 × 256	1250 × 1000	3000 × 2400

region and contains 36 ship targets. The Sentinel-1 image in Fig. 9(a) contains 112 ship targets [48]. The ground truths of the real SAR images are manually labeled according to the information provided by Wang et al. [47] and Zhang et al. [48] and with reference to Google Earth.

1) Real SAR Image 1: From the detection results of the three algorithms in Fig. 6, we can find that the algorithm proposed in this article has better detection results, as shown in Fig. 6(g) and (h). When the superpixel size is 6, the LCFV-based detector [2] fails to detect the ship target. The detected target area of the LCFV-based detector [2] is small when the superpixel size is 16. And when the superpixel size is 24, two areas are detected by the LCFV-based detector [2]. BTS-RCFAR [13] fails to detect the ship target, as shown in Fig. 6(l).

As shown in Fig. 6(i) and (j), due to the limitation of comparing adjacent superpixels, the detection results of the LCFV-based detector [2] are poor when the superpixel size is small relative to the target. In the proposed algorithm, since the features of the superpixels are obtained by comparing the superpixels with adaptive distance, the soft detection result is better than the detection result of the LCFV-based detector [2] when the superpixel size is 6. Due to the limitation of superpixel segmentation, the target detection result of the LCFV-based detector [2] consists of two areas when the superpixel size is 24. And as Fig. 6(c)–(f) shows, the soft detection results of the proposed algorithm in each level are not very good. However, a good detection result is obtained by fusing the soft detection results. BTS-RCFAR [13] does not establish an accurate background clutter statistical model of Fig. 6(a), so it fails to detect the target.

In Table IV, we quantitatively compare the detection performance of the proposed algorithm, LCFV-based detector [2], and BTS-RCFAR [13] in Fig. 6. It can be seen from Table IV that P_{dt} of the proposed algorithm is larger than the two competitive algorithms, and P_{fa} is less than the competitors. In terms of overall performance, $F1$ score of the proposed algorithm is larger than the competitors, so the overall performance of the proposed algorithm is better than the competitors. And when the superpixel size is different, P_{dt} of the LCFV-based detector [2] changes dramatically than that of the proposed algorithm, which indicates that the proposed

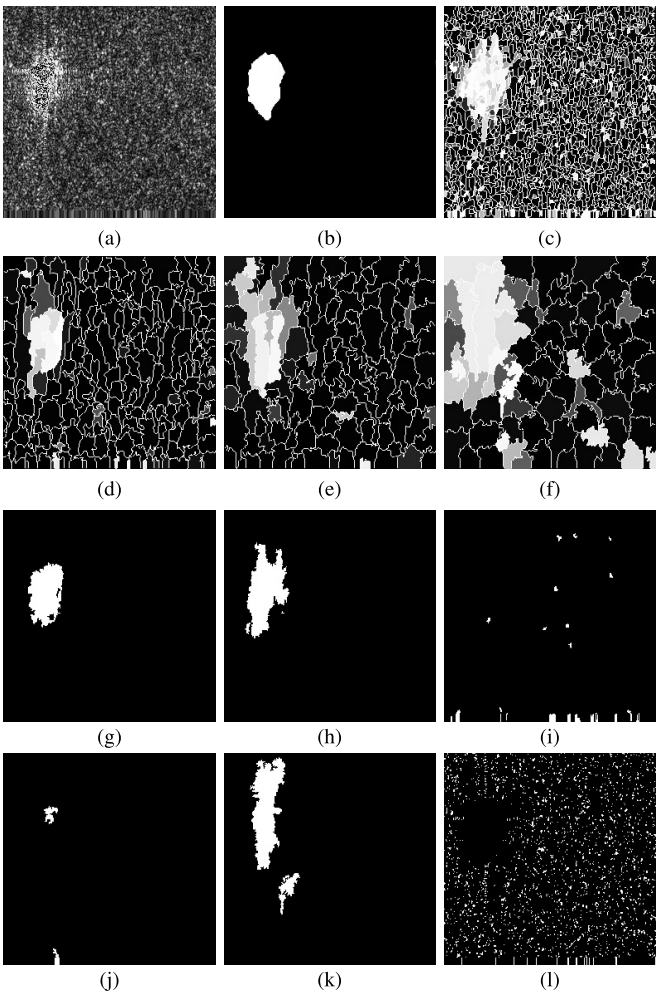


Fig. 6. (a) Gaofen-3 image with a ship target. (b) Ground truth. (c)–(f) Soft detection results of the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24, respectively. (g) Detection result of the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24. (h) Detection result of the proposed algorithm when the superpixel sizes are 8, 14, and 20. (i)–(k) Detection results of the LCFV-based detector [2] when the superpixel sizes are 6, 16, and 24, respectively. (l) Detection result of BTS-RCFAR [13].

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TABLE IV
COMPARISON OF THE SHIP TARGET DETECTION PERFORMANCE OF THREE ALGORITHMS IN FIG. 6

Algorithms	Superpixel Sizes	Fig. 6(a)		
		P_{dt}	P_{fa}	$F1$
The Proposed Algorithm	6, 12, 18, 24	0.8191	**0.1123**	**0.8520**
	8, 14, 20	**0.8780**	0.3154	0.7693
LCFV-Based Detector [2]	6	0.0076	0.9663	0.1241
	16	0.0712	0.3358	0.1287
	24	0.7027	0.5297	0.5635
BTS-RCFAR [13]	-	0.0040	0.9976	0.0030

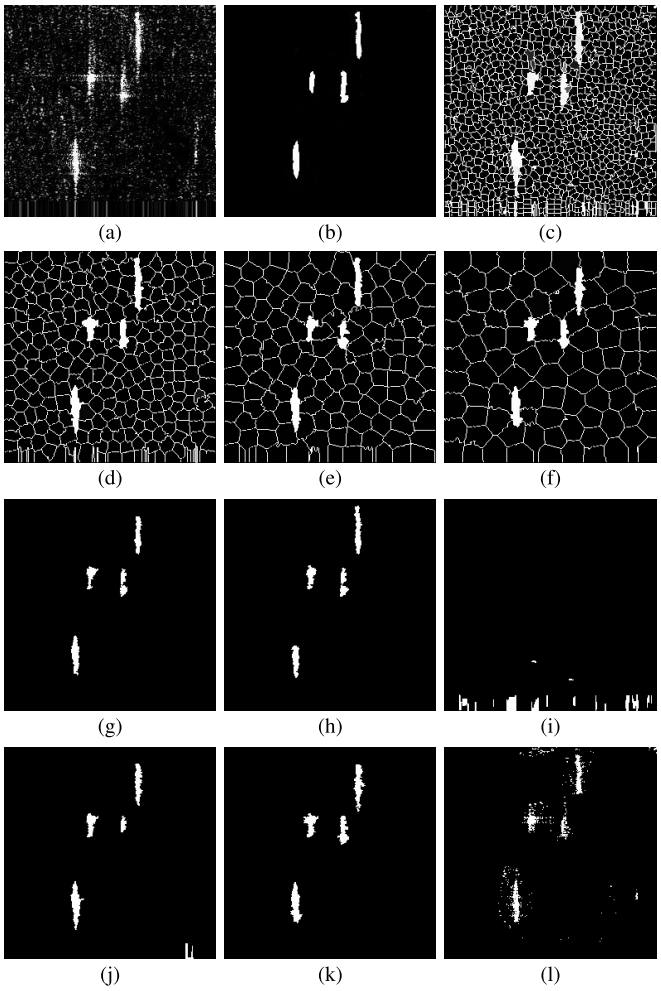


Fig. 7. (a) Gaofen-3 image with four ship targets. (b) Ground truth. (c)–(f) Soft detection results of the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24, respectively. (g) Detection result of the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24. (h) Detection result of the proposed algorithm when the superpixel sizes are 8, 14, and 20. (i)–(k) Detection results of the LCFV-based detector [2] when the superpixel size are 6, 16, and 24, respectively. (l) Detection result of BTS-RCFAR [13].

algorithm effectively weakens the influence of superpixel size selection.

2) Real SAR Image 2: From the ship detection results, as shown in Fig. 7, we can see that all the three algorithms can detect the ship targets well. However, the LCFV-based detector [2] fails to detect the targets when the superpixel size is 6. The reason is the same as before. Making comparisons of adjacent superpixels to obtain features, the features of the superpixels are not obvious. And as shown in Fig. 7(c)–(f), the proposed algorithm detects all the ship targets in each level when the superpixel sizes are 6, 12, 18, and 24 due to the adaptive distance calculation within the level. And using the multilevel fusion to eliminate false alarms in the low-level soft detection results, a good detection result is obtained. LCFV-based detector [2] has good detection results when the superpixel size is 16 or 24. BTS-RCFAR [13] also detects the targets, but many false alarms are mixed in the detection result, which requires further processing, as shown in Fig. 7(l).

In Table V, we compare P_{dt}, P_{fa}, and $F1$ score of the detection results detected by the proposed algorithm, LCFV-based detector [2], and BTS-RCFAR [13] in Fig. 7. Observing Table V, we can find that P_{dt} and P_{fa} of the proposed algorithm are better than BTS-RCFAR [13]. Although P_{dt} of the LCFV-based detector [2] is the largest when the superpixel size is 24, P_{fa} of the proposed algorithm is lower than the LCFV-based detector [2]. When the superpixel size is 6, the LCFV-based detector [2] has the smallest P_{dt} and the largest P_{fa}. $F1$ score of the proposed algorithm is higher than the two competitive algorithms, i.e., the overall performance of the proposed algorithm is better than the competitors. With the change in superpixel size, the change in P_{dt} of the proposed algorithm is smaller than that of the LCFV-based detector [2], which proves that the proposed algorithm weakens the influence of superpixel size selection.

Here, taking a look at the experimental results obtained from Figs. 6(a) and 7(a), it seems that the first superpixel size combination is more suitable for Fig. 6(a) and the second superpixel size combination is better on Fig. 7(a). Actually, this may be just occasional; if we choose different superpixel sizes, the performance may be different. But from the experimental results, we can see that both the superpixel size combinations can achieve better and almost equivalent performance with respect to the comparative algorithms. In other words, fusion makes the proposed algorithm insensitive to the choice of the superpixel size.

3) Real SAR Image 3: As shown in Fig. 8(a), the TerraSAR-X image contains 36 ship targets, and the sizes of the ship targets fluctuate strongly. The largest ship target is about 130 times of the smallest ship target. In Fig. 8(c)–(h), the correctly detected targets, missed targets, and false alarms are marked by the green boxes, yellow dotted boxes, and red ellipses, respectively.

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TABLE V
COMPARISON OF THE SHIP TARGET DETECTION PERFORMANCE OF THREE ALGORITHMS IN FIG. 7

Algorithms	Superpixel Sizes	Fig. 7(a)		
		P_{dt}	P_{fa}	$F1$
The Proposed Algorithm	6, 12, 18, 24	0.7349	**0.2296**	0.7522
	8, 14, 20	0.7697	0.2327	**0.7685**
LCFV-Based Detector [2]	6	0.0	1.0	-
	16	0.7124	0.3735	0.6667
	24	**0.8301**	0.3528	0.7274
BTS-RCFAR [13]	-	0.6602	0.4678	0.5893

As can be seen, the proposed algorithm detects 36 correct targets, 0 missed target, and 0 false alarm when the superpixel sizes are 6, 12, 18 and 24, as shown in Fig. 8(c). As Fig. 8(d) shows, when the superpixel sizes are 8, 14, and 20, the proposed algorithm also detects all the ships. As for the competitors, the superpixel size has a great impact on the detection result when the LCFV-based detector [2] is adopted to detect the targets. As shown in Fig. 8(e), the LCFV-based detector [2] misses many ship targets and has many false alarms when the superpixel size is 6. As shown in Fig. 8(f) and (g), when the superpixel size is 16 or 24, the LCFV-based detector [2] correctly detects 33 and 27 ships, respectively. And there are many false alarms when the superpixel size is 16. In the LCFV-based detector [2], the features of the superpixels are obtained by comparing adjacent superpixels; when the sizes of the ship targets fluctuate greatly, it is difficult to obtain the superpixel features for large and small targets at the same time. BTS-RCFAR [13] detects all the ship targets, as shown in Fig. 8(h), but there are also a lot of false alarms at the same time. From the detection results, it can be seen that the proposed algorithm can weaken the influence of the superpixel size selection. Besides, due to the power of fusion, the proposed algorithm is insensitive to the size of the ship targets. Or better, the proposed algorithm suits the situation when the image contains multiple targets with different sizes.

The quantitative comparison of the detection performance in Fig. 8 is shown in Table VI. We can find that although P_{dt} of the three algorithms is similar, $F1$ score of the proposed algorithm is much larger than the competitors due to the extremely low P_{fa}. The overall performance of the proposed algorithm is better than the two competitive algorithms. And the detection performance of the LCFV-based detector [2] changes more severe than that of the proposed algorithm when the superpixel size is different, which further proves that the proposed algorithm weakens the influence of superpixel size selection.

4) Real SAR Image 4: As shown in Fig. 9(a), the Sentinel-1 image contains 112 ship targets, and the largest ship target size is about 30 times of the smallest ship target size.

As shown in Fig. 9(c), the proposed algorithm detects 100 correct targets, 12 missed targets, and 0 false alarm when the superpixel sizes are 6, 12, 18. and 24. And as Fig. 9(d) shows, the proposed algorithm detects 102 correct targets, 10 missed targets, and 0 false alarm when the superpixel sizes are 8, 14, and 20. Satisfying performance can be achieved using the proposed algorithm. When adopting the LCFV-based detector [2] to detect the targets, the superpixel size has a great impact on the detection result. As shown in Fig. 9(e) and (g), when the superpixel size is 6 or 24, the LCFV-based detector [2] only correctly detects 73 and 75 targets, respectively. And the reason for poor detection performance is the same as the previous experimental analysis. When the superpixel is 16, as shown in Fig. 9(f), the detection result of the LCFV-based detector [2] includes 102 correctly detected targets, ten missed targets, and 22 false alarms. The choice of the superpixel size has a great impact on the final detection performance. As shown in Fig. 9(h), BTS-RCFAR [13] detects 110 correct targets and two missed targets, but it still detects quite a lot of false alarms. From the detection results, we can find that the proposed algorithm is insensitive to the superpixel size. Satisfying detection results can be achieved.

P_{dt}, P_{fa}, and $F1$ score of the detection results detected by the proposed algorithm, LCFV-based detector [2], and BTS-RCFAR [13] in Fig. 9 are listed in Table VII. Although P_{dt} of the proposed algorithm is not the best, P_{fa} is the smallest because of multilevel fusion, which makes $F1$ score far better than the two competitive algorithms. Due to the fluctuation of the ship target size, the performance of the LCFV-based detector [2] is not satisfactory, and the influence of the selection of superpixel size on the detection results is obvious. The proposed algorithm eases the influence of superpixel size selection and obtains the best overall detection performance.

For the comparative LCFV-based detector [2], the choice of the superpixel size has an important impact on the target detection results. The better results for Figs. 8(a) and 9(a) can be obtained by setting more suitable superpixel sizes. And more postprocessing can also be implemented to remove false alarms. For the experimental results of BTS-RCFAR [13] obtained from Figs. 8(a) and 9(a), we only use the morphological open operation with a kernel size of 4 as the postprocessing step. And in fact, more precise postprocessing steps will lead to a better detection result.

It can be concluded from the above experimental results that the proposed algorithm is effective and superior in accurately detecting ship targets and reducing false alarms. And the proposed algorithm weakens the influence of superpixel size selection, even when the sizes of ship targets fluctuate strongly.

C. Complexity Analysis and Running Time

In this section, we analyze the complexity of the three algorithms at first. And then we compare the running time of BTS-RCFAR [13], LCFV-based detector [2], and the proposed algorithm. The hardware environment of the experiments is

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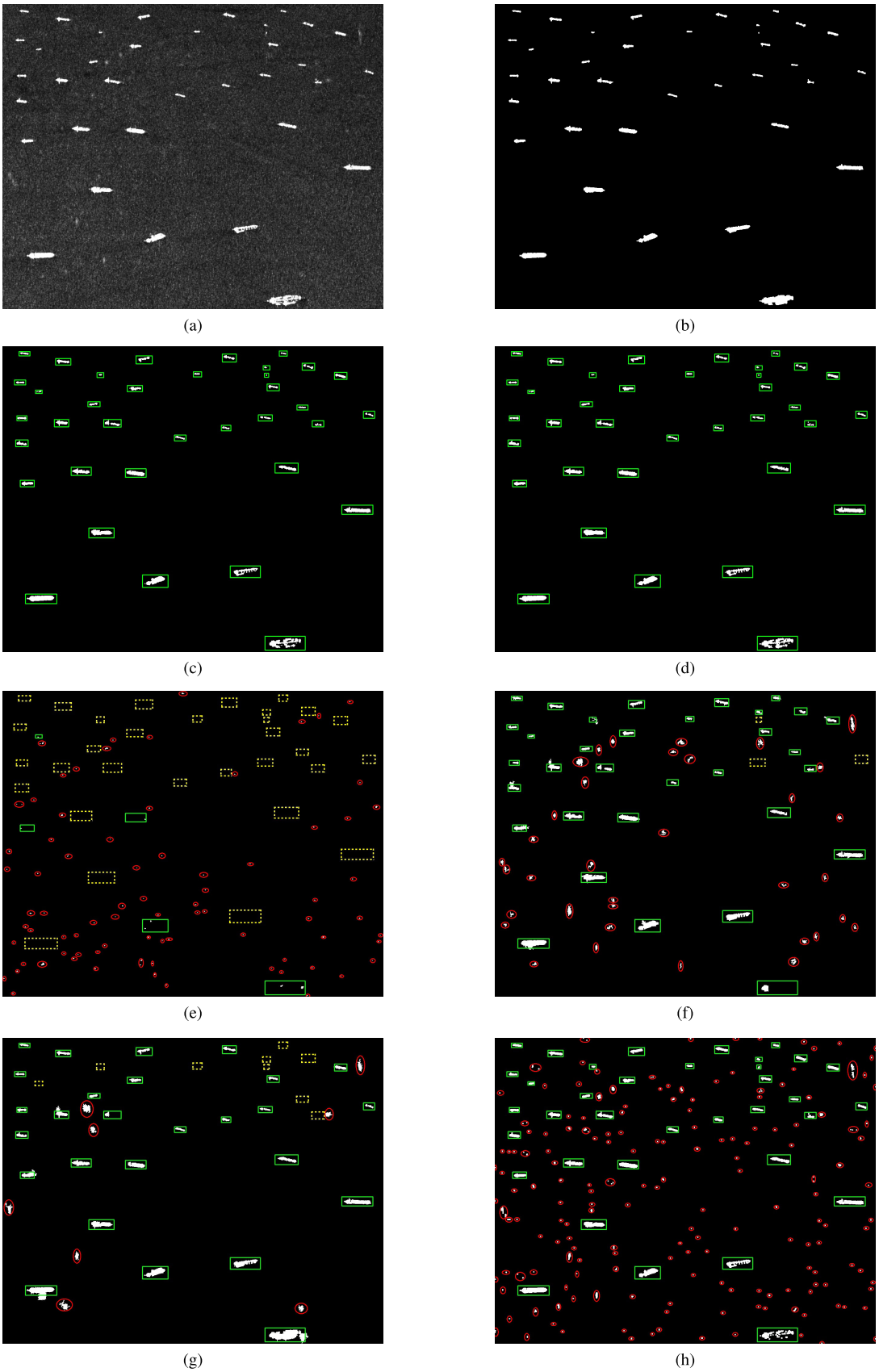


Fig. 8. (a) TerraSAR-X image with 36 ship targets. (b) Ground truth. (c) Detection result of the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24. (d) Detection result of the proposed algorithm when the superpixel sizes are 8, 14, and 20. (e)–(g) Detection results of the LCFV-based detector [2] when the superpixel sizes are 6, 16, and 24, respectively. (h) Detection result of BTS-RCFAR [13].

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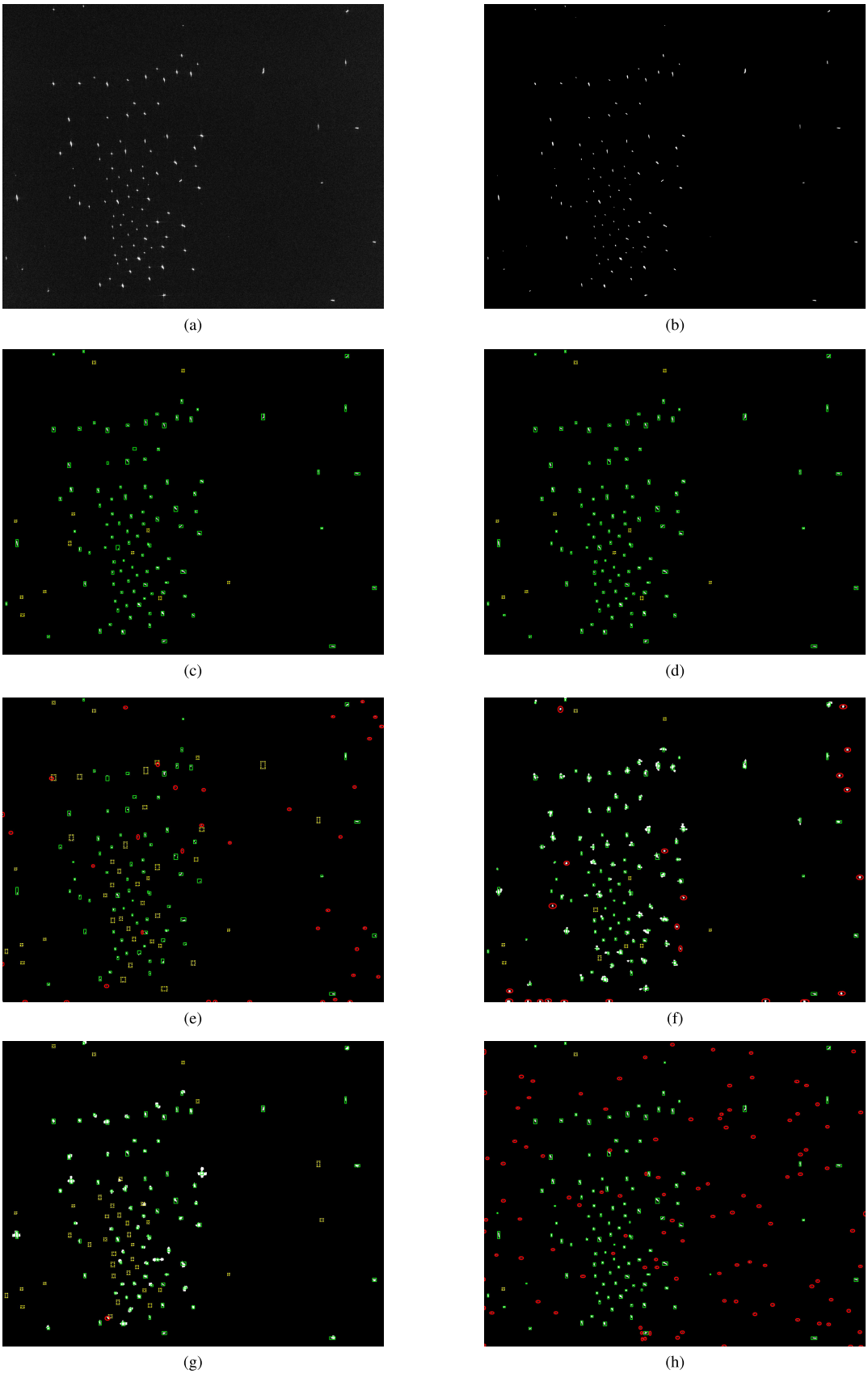


Fig. 9. (a) Sentinel-1 image with 112 ship targets. (b) Ground truth. (c) Detection result of the proposed algorithm when the superpixel sizes are 6, 12, 18, and 24. (d) Detection result of the proposed algorithm when the superpixel sizes are 8, 14, and 20. (e)–(g) Detection results of the LCFV-based detector [2] when the superpixel sizes are 6, 16, and 24, respectively. (h) Detection result of BTS-RCFAR [13].

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TABLE VI
COMPARISON OF THE SHIP TARGET DETECTION PERFORMANCE OF THREE ALGORITHMS IN FIG. 8

Algorithms	Superpixel Sizes	Fig. 8(a)		
		P_{dt}	P_{fa}	$F1$
The Proposed Algorithm	6, 12, 18, 24	0.8646	**0.0083**	0.9238
	8, 14, 20	**0.8877**	0.0153	**0.9337**
LCFV-Based Detector [2]	6	0.0037	0.9508	0.0069
	16	0.7741	0.3780	0.6898
	24	0.8873	0.2840	0.7925
BTS-RCFAR [13]	-	0.8480	0.3742	0.7201

TABLE VII
COMPARISON OF THE SHIP TARGET DETECTION PERFORMANCE OF THREE ALGORITHMS IN FIG. 9

Algorithms	Superpixel Sizes	Fig. 9(a)		
		P_{dt}	P_{fa}	$F1$
The Proposed Algorithm	6, 12, 18, 24	0.9006	**0.0196**	**0.9388**
	8, 14, 20	0.9355	0.0688	0.9334
LCFV-Based Detector [2]	6	0.3288	0.4819	0.4023
	16	**0.9585**	0.8284	0.2910
	24	0.8336	0.7911	0.3340
BTS-RCFAR [13]	-	0.9497	0.4154	0.7237

Intel[1] Core[2] i5-6600 CPU at 3.30 GHz and 8-GB RAM, and the software environment is Python 3.8.

1) Complexity Analysis: The complexity of BTS-RCFAR [13] is $O(W^2N)$, where W is the size of the clutter window. The complexity of the LCFV-based detector [2] is mainly composed of four parts. The first part is the complexity of parameter estimation of the Gaussian mixture model using expectation–maximization (EM) algorithm, which is $O(GIN)$, where G is the number of Gaussian distributions in the Gaussian mixture model and I is the number of iterations. The second part is the complexity of the superpixel segmentation by SLIC, which is $O(N)$ [27]. The third part is the complexity of calculating with the contrast superpixels, which is $O(BL)$, where B is the number of the contrast superpixels. The fourth part is the complexity of processing superpixels, which is $O(L)$. So the complexity of the LCFV-based detector [2] is $O(GIN)$.

The complexity of the proposed algorithm is also mainly composed of four parts. The first part is the complexity of SLIC [27] for superpixel segmentation at different levels, which is $O(\tau N)$, the second part is the complexity of calculating with the contrast superpixels, which is $O(\tau BL \cdot S^2 \log S)$ that equals $O(\tau BN \log S)$, the third part is the complexity of the FCM algorithm, which is $O(\tau cIL)$, and the fourth part is the complexity of processing pixels, which is $O(N)$. So, the complexity of the proposed algorithm in this article is $O(\tau BN \log S)$. We can see that the complexity of all the three algorithms is related to $O(N)$.

2) Running Time: In Table VIII, the detection time of the LCFV-based detector [2] in Figs. 6(a) and 7(a) is the time with superpixel size 24. The time of the proposed algorithm in Figs. 6(a) and 7(a) is the time of four-level fusion detection with superpixel sizes of 6, 12, 18, and 24.

[1]Registered trademark.
[2]Trademarked.

TABLE VIII
COMPARISON OF RUNNING TIME OF THREE ALGORITHMS

Time(s) SAR Images / Algorithms	Fig. 6(a)	Fig. 7(a)
BTS-RCFAR [13]	114	109
LCFV-Based Detector [2]	18	18
The Proposed Algorithm	**8**	**5**

It can be seen from Table VIII that the detection time of the proposed algorithm and LCFV-based detector [2] is far less than that of BTS-RCFAR [13]. This shows that the complexity coefficient of BTS-RCFAR [13] is much greater than that of the LCFV-based detector [2] and the proposed algorithm in practical use. Moreover, we can conclude that the use of superpixels greatly reduces the time of detection in SAR images. And we can see from Table VIII that the detection time of the proposed algorithm is the shortest, i.e., the complexity coefficient of the proposed algorithm is the smallest.

V. CONCLUSION

In this article, we propose a new SAR image ship detection algorithm based on multilevel superpixel segmentation and fuzzy fusion. In the proposed algorithm, we first perform superpixel segmentation at different levels of the SAR image using different superpixel sizes, and then adaptively calculate and select the w_1 distance between superpixels within each level to perform soft detection, and finally fuse the soft detection results to achieve the ship target detection result. In the proposed algorithm, we establish a new method of adaptively selecting the distance between superpixels according to the size of superpixels when calculating the features of superpixels, and construct a new fusion strategy of pixel-level clutter probability between different levels. The experimental results show that the proposed algorithm finally achieves the purpose of improving the performance of ship target detection

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and weakening the influence of the superpixel size selection on detection performance.

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